Paleomagnetism of the Tulemalu dykes, Northwest Territories, Canada

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Easterly trending Tulemalu diabase dykes that cut an area of predominantly Archean rock in southern Keewatin were magnetized more than 2200 Ma ago. Thirteen of the sampled sites (dykes) exhibit a southeast-down remanent magnetization and five sites give a more poorly grouped direction that is reversed to this. Their combined pole is at 122.4°E, 0.5°S, δm = 10.4°, δp = 6.0°. Six northeasterly trending Kazan dykes, which are probably correlatives of the Kaminak dykes, give a metamorphic pole at 94.2°W, 30.7°N, δm = 35°, δp = 31°. The age of their magnetization is thought to be between 1900 and 2000 Ma.

The pole for the Tulemalu dykes lies within a group of poles of similar age derived from basic rocks of the Slave Province. It is about 30° away from the pole for the MacKay dykes, which occur north of Great Slave Lake and which have an easterly trend similar to that of the Tulemalu.

Poles whose ages range between 1900 and 2200 Ma and that are derived from the Slave, western Churchill, and Superior provinces all suggest that during this period these three continental plates occupied approximately the same relative positions as they do today.

Les dykes de diabase de Tulemalu de direction est, lesquels coupent une région constituée principalement de roches de l'Archéen dans le Keewatin sud, ont acquis leur aimantation il y a plus de 2200 Ma. Treize des sites d'échantillonnage (dykes) donnent une aimantation rémanente en bas vers le sud-est et cinq sites fournissent une direction moins uniformément groupée laquelle est inversée par rapport à la première. Leur pôle combiné est à 122,4°E, 0,5°S, δm = 10,4°, δp = 6,0°. Six dykes de Kazan de direction nord-est sont probablement en corrélation avec les dykes de Kaminak qui fournissent un pôle métamorphique à 94,2°O, 30,7°N, δm = 35°, δp = 31°. L'âge de leur aimantation remonte probablement entre 1900 et 2000 Ma.

Le pôle des dykes de Tulemalu se situe à l'intérieur d'un groupe de pôles d'âge analogue obtenu sur des roches basiques de la province des Esclaves. Il dévie d'environ 30° du pôle des dykes de MacKay qui apparaissent au nord du grand lac des Esclaves et avec une direction est analogue à celle de Tulemalu.

Les pôles dont les âges sont compris entre 1900 et 2200 Ma et qui dérivent des provinces des Esclaves, de Churchill ouest et du Supérieur indiquent tous que durant cette période ces trois plaques continentales occupaient approximativement les mêmes positions relatives que celles d'aujourd'hui.

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Geology

The Tulemalu dykes make up a fairly widespread swarm centred on the Tulemalu map area of southern Keewatin. They extend into, but are less abundant in, the northern half of the Ferguson Lake map area to the east (Fig. 1), and a few dykes of similar character have been reported both to the north and south of the Tulemalu Lake map area (Eade 1984). The predominant trend of the Tulemalu dykes is easterly to east-southeasterly, and radioisotopic dating and field relationships indicate that they are early Aphebian (see Eade and Wanless in Wanless *et al.* 1980). Eade (1984) has indicated that some easterly trending members of the swarm were emplaced in fault zones and that later movement on the faults has imparted intense cleavage to some of these dykes. Most of the dykes, however, do not have trends that parallel the known regional faults so they probably reflect a tensional regime differing from that existing during the formation of the fault systems.

The Tulemalu dykes are not identifiable on standard Geological Survey of Canada 1:50 000 aeromagnetic maps so most of the dykes shown in Fig. 2 were identified by direct ground or air observation. Dykes do seem to be apparent on more detailed aeromagnetic maps, and the pattern within the rectangle outlined by dots in Fig. 2 was derived from 1:20 000 maps loaned by Pan Ocean Oil Limited. Linear anomalies on these maps agree well with dykes as mapped on the ground so it is very likely that they indicate the presence of dykes.

Metamorphic alteration of silicate minerals in Tulemalu material from sites sampled for paleomagnetic study varies from extremely slight to moderate. The alteration takes the form of slight chloritization of mafic minerals in least altered dykes to development of amphibole coronas around pyroxenes, or selective replacement of pyroxene by amphibole and chlorite. Epidote veinlets are present in some of the outcrops sampled. These metamorphic features appear to be the results of low-grade regional metamorphism whose affect is irregularly distributed within Tulemalu map area. According to Eade (1984) this alteration is more pervasive in Tulemalu dykes of the Ferguson Lake map area just to the east.

Besides the Tulemalu dykes, which are the chief interest of the present paper, Eade (1984) has described a set of northeasterly trending metagabbro dykes in the Tulemalu map area. These are the Kazan dykes, which Eade suggested are Archean in age. Six of these dykes were sampled for paleomagnetic study. Only these six Kazan dykes are shown in Fig. 2 in order to avoid further clutter to an already complex diagram. Sampling of the Kazan dykes was undertaken to provide paleomagnetic results from rocks of the predominantly metamorphic terrane that forms the regional background to the Tulemalu intrusions.

A few northwesterly trending Mackenzie dykes (1.2 Ga), also omitted from Fig. 2, are present in the Tulemalu Lake map area. These have a strong signature on standard 1:50 000 aeromagnetic maps.

The age of the Tulemalu dykes and their magnetization

A concerted attempt has been made in recent years to determine the ages of the various rock units that occur within the Tulemalu Lake and adjoining map areas. The results of this

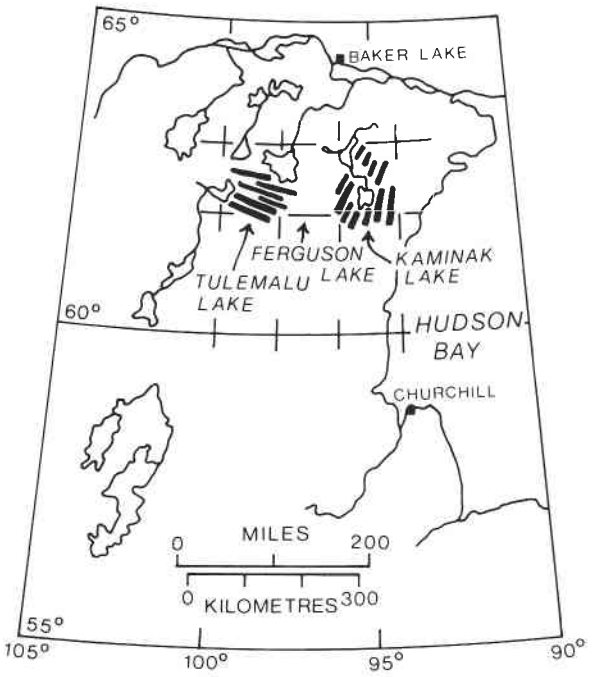


FIG. 1. Sketch showing the locations of the Tulemalu, Ferguson, and Kaminak map areas and the location and trends of the two major dyke swarms of the region.

dating program, which includes work on the Tulemalu dykes (Table 1, Fig. 2), have been discussed by Eade and Wanless in Wanless *et al.* (1980) and also by Eade *et al.* (1984). The oldest rocks of the area are felsic tuffs, granodiorites, granodiorite orthogneiss, and quartz monzonite. Material from these rocks yielded U–Pb zircon ages clustering around 2650 Ma. These may be primary ages, and using Stockwell's (1982) terminology the intrusions and their extrusive equivalents formed during the Laurentian Orogeny, which is late middle Archean.

Three K–Ar analyses of hornblendes from the Archean granodiorite and quartz monzonite country rocks have yielded ages of about 2450 Ma. These are interpreted as indicating the time when the hornblendes had cooled to a temperature below which they no longer spontaneously lost radiogenic argon. Four hornblendes from country rock in contact with Tulemalu dykes were also analyzed. Two of these yielded K–Ar ages of 2471 ± 62 and 2473 ± 184 Ma (Table 1), which are identical, within limits of error, to the three country rock hornblende ages of 2485 ± 62, 2431 ± 61, and 2429 ± 209 Ma. This is not surprising because hornblende can be very resistant to argon loss in response to short-lived thermal events of this type. The dykes with which these hornblendes were in contact were only 16 m thick. The other two contact hornblendes yielded ages of 2250 Ma. These were in contact with thicker dykes (22 and 25 m) and as it seems unlikely that the times at which country rock hornblendes reached their blocking temperatures differed by 200 Ma as a result of differential uplift these two hornblende ages probably reflect overprinting by the intrusion of the Tulemalu dykes.

A single K–Ar age of 2100 Ma was obtained from biotite in contact with a Tulemalu dyke. As would be expected, this age is less than the contact hornblende age from the Tulemalu dyke, probably reflecting a more complete loss of argon. Six whole-rock chilled dyke margin analyses yielded a mean age of 2086 Ma, similar to that of the contact biotite, 2118 Ma. This confirms previous experience indicating that unaltered chilled whole-rock diabase dyke ages are similar to contact country rock biotite ages. The three youngest whole-rock dyke margin ages may reflect mild reheating at about 2000 Ma.

The fact that country rock hornblendes are oldest at 2450 Ma, country rock biotites slightly younger at 2325 Ma, two contact hornblendes still younger at 2250 Ma, and chilled whole rocks and a contact biotite the youngest at 2100 Ma suggests that we are not observing simply a blanket regional metamorphism at about 2450 Ma. Rather, the data suggest an older metamorphism on which has been imposed later local thermal effects related to the Tulemalu dyke intrusions. An age of Tulemalu intrusion close to the 2200 Ma suggested by Eade (1984) therefore seems quite reasonable.

Previous work relating paleomagnetism and K–Ar ages of diabase dyke swarms indicates that the paleomagnetic signature is at least as old as the chilled whole-rock ages. For this reason it is suggested that the dominant magnetization is a thermal remanent magnetization (TRM) developed at or near the time of the dyke intrusion some 2200 Ma ago.

Potassium–argon ages from the older metamorphosed Kazan dykes were somewhat surprising. Whole-rock margin, fine-grained marginal hornblende, and contact biotite were analyzed from each of two Kazan dykes. Most of the potassium in the "chilled" margins probably resides in very fine-grained hornblende so the whole-rock analyses are essentially equivalent to the fine-grained hornblende analyses. In any case the six analyses gave rather coherent results suggesting a K–Ar age of about 1900 Ma. This may be indicative of the Blezardian Orogeny (Stockwell 1982); in support of this, we note that hornblende from a younger quartz monzonite intrusion in the northwest part of the Tulemalu area (Fig. 2) gave a K–Ar age of 1984 ± 78 Ma. Inasmuch as metagabbro Kazan dykes occur elsewhere in the Tulemalu map area near unaltered Tulemalu dykes these younger ages suggest that the southwest part of the area was affected by Blezardian metamorphism, which had little or no effects to the east. If such is the case these Kazan dykes were probably metamorphosed at least twice: once prior to, and once after, the intrusion of the Tulemalu swarm.

Sampling and laboratory treatment

One hundred and sixty-nine cores were drilled and sun oriented at 25 sites in the Tulemalu Lake map area. One hundred and sixty-five were from dykes; the remaining four were from rock in contact with dykes. Nineteen of the dykes sampled (129 cores) have an easterly or east-southeasterly Tulemalu trend; the remaining six dykes (40 cores) have a northeasterly Kazan trend. From each core at least two cylindrical specimens were cut and alternating field (AF) demagnetized in 8–10 steps. At lower fields a home-built tumbler demagnetizer and biastatic magnetometer were used; at higher fields a Schonstedt demagnetizer and spinner magnetometer were used. In addition to the AF treatment, generally two specimens from each site were thermally demagnetized.

Laboratory results

Tulemalu dykes

Seventeen of the 19 Tulemalu sites gave coherent results and most had optimum cleaning fields between 40 and 100 mT. In instances in which major intensity drops occurred with no significant change in direction or dispersion over a range of fields, the data for the highest field are generally given in Table 2 and indicated in Fig. 3. In Fig. 2, the cleaned site directions are

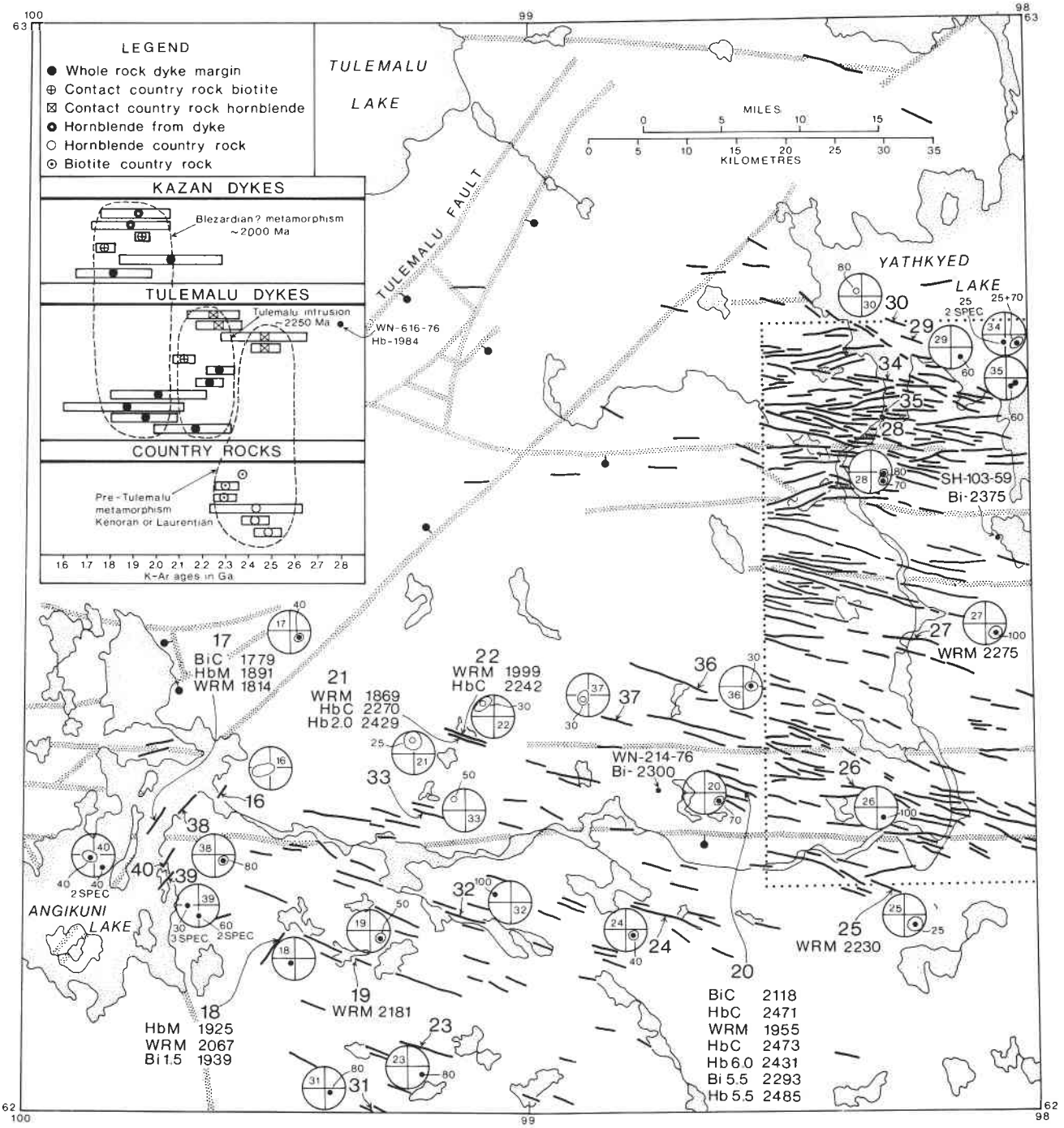


FIG. 2. K–Ar ages and paleomagnetic data from dykes of Tulemalu area. Numbers within stereonets are site numbers as are large numbers that are connected by arrows to collecting sites. Numbers and letters below large site numbers indicate K–Ar ages (Ma) (see Table 2) and the material dated: HbM = hornblende from dyke margin; WRM = chilled whole-rock margin, Bi 1.5 = country rock biotite 1.5 m from dyke margin; Hb 6.0 = country rock hornblende 6.0 m from dyke margin; Bi = country rock biotite; HbC = country rock contact hornblende; BiC = country rock contact biotite. Heavy stippled lines are fault zones; downthrown side is indicated. Mean magnetization directions are shown as dots (down) or circles (up) and lines around dots indicate diagrammatically the dispersion around site mean directions. Numbers outside the stereonets and connected to the dots (circles) indicate optimum cleaning fields in milliteslas. The designation 2 SPEC beneath an AF field indicates that two specimens gave this direction at the given field.

shown as dots and circles and dispersion is indicated diagrammatically, with the area enclosed by a line suggesting the degree of dispersion. Orthogonal plots indicate that two stable components were isolated from the material of several sites. At some sites the two components are present in all specimens of the site (sites 28 and 35, Fig. 2). In others, some specimens from the site yielded one component and the remaining cores yielded another (site 34, Fig. 2). Analysis of step-cleaning data from most Tulemalu specimens, however, showed evidence of only a steep, viscous, recent Earth's field component, along with one other, either primary TRM or metamorphic component.

Twelve of the Tulemalu sites yielded a stable, moderately southeast-down direction and five gave a poorly grouped, ap-

TABLE 1. Potassium argon and U−Pb ages

Paleomagnetic site	GSC No.	Sample No.	Method	Age (Ma)
FA-7617	78-166	WN-216-76	Biotite from country rock in contact with 30 m wide metadiabase dyke	1779± 42
	78-162	WN-215-76	Hornblende from chilled margin of 30 m wide metadiabase dyke	1891±169
	78-163	WN-215-76	Whole rock from chilled margin of 30 m wide metadiabase dyke	1814±165
FA-7618	78-164	WN-217-76	Very fine-grained hornblende from margin of 60 m wide metadiabase dyke	1925±157
	78-167	WN-218-76	Biotite from country rock 1.5 m from dyke contact	1939± 34
	78-165	WN-217-76	Whole rock from chilled margin of 60 m wide dyke	2067±226
FA-7619	78-146	WN-219-76	Whole rock from chilled margin of 25 m wide dyke	2181±152*
FA-7620	78-153	EA-1329-75	Biotite from country rock in contact with 16 m wide dyke	2118± 47
	78-154	EA-1329-75	Hornblende from country rock in contact with 16 m wide dyke	2471± 62
	78-152	EA-400-75	Whole rock from chilled margin of 16 m wide dyke	1955±106
	78-157	EA-1328-75	Hornblende from country rock 5.5 m from dyke contact	2485± 62
	78-159	WN-212-76	Hornblende from country rock in contact with 16 m wide dyke	2473±184
	Unpublished	WN-212-76	Zircon (U−Pb) near dyke	2671
	78-155	EA-1327-75	Hornblende from country rock 6 m from 16 m wide dyke	2431± 61
	78-156	EA-1328-75	Biotite from country rock 5.5 m from 16 m wide dyke	2293± 51
FA-7621	78-147	WN-221-76	Whole rock from chilled margin of 25 m wide dyke	1869±220
	78-149	WN-222-76	Hornblende from country rock in contact with 25 m wide dyke	2270±100
	78-150	WN-223-76	Hornblende from country rock 2 m from 25 m wide dyke	2429±203
	Unpublished	WN-223-76	Zircon (U−Pb) near dyke	2659
FA-7622	78-148	WN-224-76	Whole rock from chilled south margin of 22 m wide dyke	1999±213
	78-151	WN-225-76	Hornblende in contact with 22 m wide dyke	2242±114
FA-7625	78-161	EA-76A-75	Whole rock from chilled margin of 30 m wide dyke	2230± 59
FA-7627	78-160	EA-65A-75	Whole rock from chilled margin of 23 m wide dyke	2275± 59
—	78-158	WN-214-76	Biotite from granodiorite	2300± 53
—	61-104	SH-103-59	Biotite from granite gneiss	2375
—	81-150	WN-616-76	Hornblende from younger quartz monzonite	1984± 78

*The original published whole-rock age was 2549±265 Ma; the value given here is the result of a rerun in 1983.

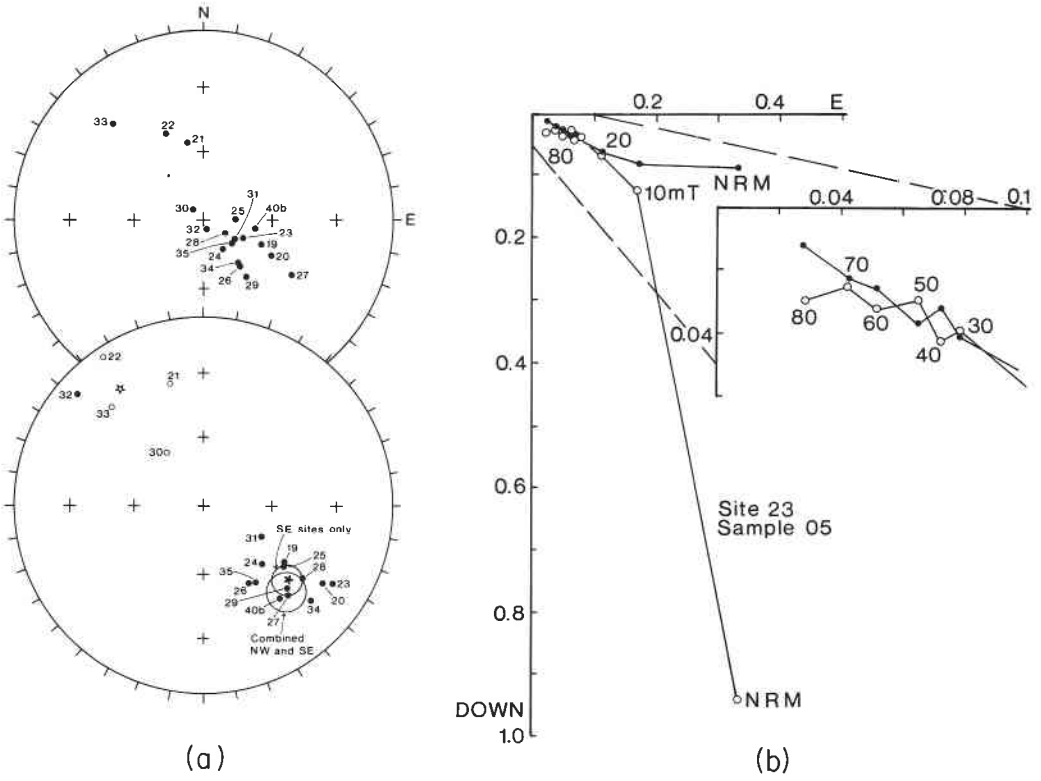


FIG. 3. (*a*) Equal-area projection of site mean directions of 18 Tulemalu dykes. NRM site mean directions are given above, cleaned site mean directions below. Solid circles = north-seeking down; open circles = north-seeking up. Open star is northwest mean up direction; solid star is southeast mean down direction. The two larger circles are circles of confidence at $P = 0.05$. Combined northwest and southeast circle of confidence includes all 19 sites with NW directions reversed. (*b*) Typical orthogonal plot of AF step demagnetization of Tulemalu dyke.

TABLE 2. Paleomagnetic data for the Tulemalu diabase dykes

			NRM				Cleaned					
Site	N	n	$D°$	$I°$	K	$\alpha_{95}°$	AF (mT)	$D°$	$I°$	P	K	$\alpha_{95}°$
						Tulemalu dykes						
19	7	6	114	62	160	5	50	125	46	R	80	7
20	5	5	118	56	414	4	70	124	25	R	74	9
21	7	5	349	55	3	54	25	345	−33	N	6	35
22	6	5	337	48	21	17	30	326	−5	N	20	17
23	7	6	116	71	73	8	80	121	21	R	44	10
24	7	5	146	75	33	14	40	135	53	R	49	11
25	8	6	89	76	30	12	25	127	45	R	24	14
26	6	6	142	64	35	11	100	149	50	R	115	6
27	6	6	122	44	4	36	100	137	34	R	7	27
28	6	5	123	79	163	6	70	126	35	R	14	22
29	6	6	144	59	54	9	60	135	37	R	26	14
30	6	6	321	83	59	9	80	326	−62	N	57	9
31	7	6	122	74	95	7	80	118	61	R	48	10
32	7	7	152	86	23	13	100	312	10	N	8	23
33	7	5	317	30	9	26	50	317	−29	N	110	7
34	6	5	142	66	Random		70	132	24	R	20	18
35	7	6	127	74	315	4	60	146	49	R	33	12
40b	2	2	100	64			40	140	36	R		
Mean direction of 5 normal sites above (cleaned)								325	−24		8	30
Mean pole position from mean direction							116.0°E	10.1°N		δm=31.5°	δp=16.9°	
Mean pole position from 5 normal site poles							115.0°E	7.9°N			14	21
Mean direction of 13 reversed sites above (cleaned)								132	40		34	7
Mean pole position from mean direction							55.1°W	3.3°N		δm= 8.7°	δp= 5.2°	
Mean pole position from 13 reversed site poles							55.6°W	4.1°N			43	6
Mean combined direction of 18 normal and reversed sites above								136	36		16	9
Mean pole position from combined mean direction							122.4°E	0.5°S		δm=10.4°	δp= 6.0°	
Mean pole position from 18 combined site poles							121.9°E	0.9°S			25	7

NOTES: N = number of cores collected at a site; n = number after eliminating those cores that were unstable; D = declination; I = inclination; K = Fisher's estimate of precision; α_{95} = semi-axis of the circle of confidence at $P = 0.05$; AF = peak alternating field; P = polarity, arbitrarily designated reversed (R) if north-seeking direction is down to the southeast; N = normal; δm, δp = semi-axes of ellipse of confidence at $P = 0.05$.

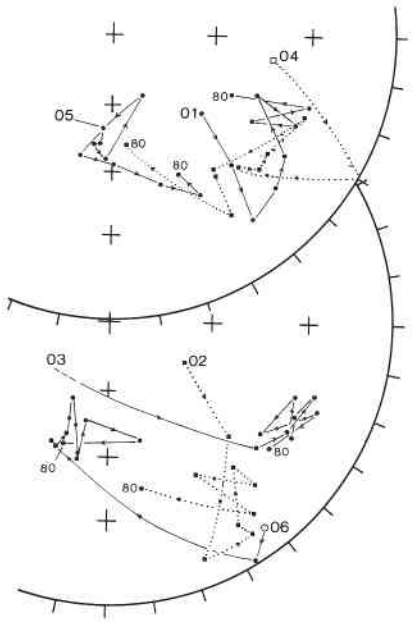


FIG. 4. Detailed AF demagnetization of six cores from site 34. Cores 05 and 06 are from the epidotized north side of the dyke; cores 01−04 are from less altered rock that yields a Tulemalu direction.

parently reversed direction (Fig. 3). Two sites do not fit this pattern. Site 37 has a steeply up southwest direction and site 36 an easterly down direction. The magnetization of these sites is not particularly stable. A stable easterly component also seems to be present in site 28 and in three specimens of site 18. A 45° clockwise rotation about a vertical axis of site 37 could bring the directions for this site into agreement with the northwest-up Tulemalu direction.

It was observed in the field that the dyke sampled at site 40 is more similar in appearance to the Tulemalu dykes than to the Kazan dykes, although it has the northeast Kazan strike. The two least altered (metamorphosed) cores from this site are very stably magnetized in a Tulemalu direction. These two samples retained this direction after having been step cleaned to 80 mT then step cleaned thermally to 560°C. The remaining five specimens exhibited a very steep southwest-down direction. The two specimens with a southeast magnetization have been included in the Tulemalu group of dykes as site 40b (Table 2). In a similar way it was observed that the degree of epidotization increases steadily from south to north across the dyke at site 34. Six cores were drilled across the dyke from south to north and the two most altered cores (5 and 6) were found to exhibit a steep south-southwest magnetization, whereas the other four cores (1−4) yielded the Tulemalu direction (Fig. 4). These data

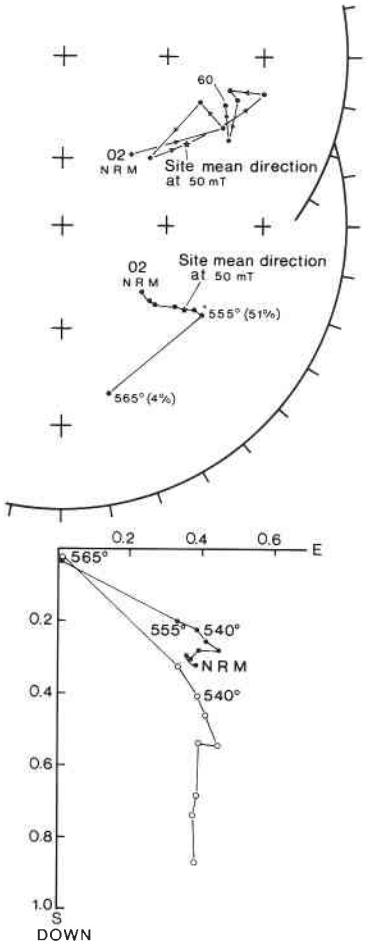


FIG. 5. Response of baked sample 1902 to AF (top) and thermal (middle) step cleaning. Lower diagram is orthogonal plot of the thermal demagnetization: closed circles are on horizontal plane, open circles are on vertical plane. Numbers with degrees indicate temperature in degrees Celsius; numbers followed by percentage symbols give percentage of initial intensity remaining after thermal cleaning at the indicated temperature.

are taken to support the suggestion that the steep south direction is later and related to a metamorphic episode.

Contact samples

Two samples were drilled from the country rock of the Tulemalu dykes at each of sites 19 and 20. One contact sample from site 19 was very weakly magnetized; however, a second sample 10 cm from the contact was quite strongly and stably magnetized. The response of this sample to thermal and AF demagnetization is shown in Fig. 5. There is evidence in this sample of only one stable component with a direction in good agreement with that of the Tulemalu dykes. An amphibolite sample 5 cm from the dyke margin at site 20 also exhibited only Tulemalu magnetization (Fig. 6). A second sample from country rock drilled 20 m from the dyke margin at site 20 contained a fairly stable, steep, southerly component well away from the present Earth's field direction and well away from the remanent direction of the dyke and of the country rock contact sample (Fig. 7). Unfortunately the more distant sample was drilled in quartz monzonite because the contact amphibolite did not extend far from the dyke margin. For this reason the contact test is not ideal but the data do suggest that the magnetization of the dyke was primary.

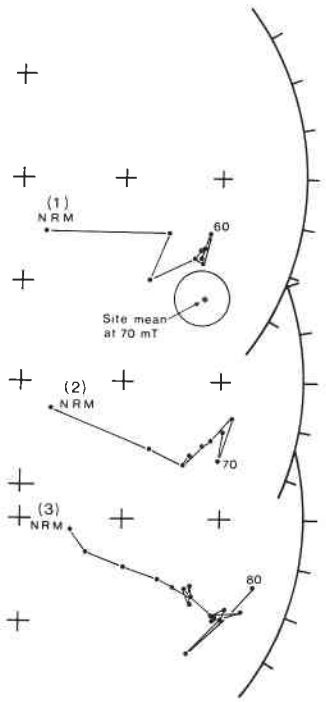


FIG. 6. Response of three specimens of baked amphibolite (sample 06, site 20) to AF step cleaning.

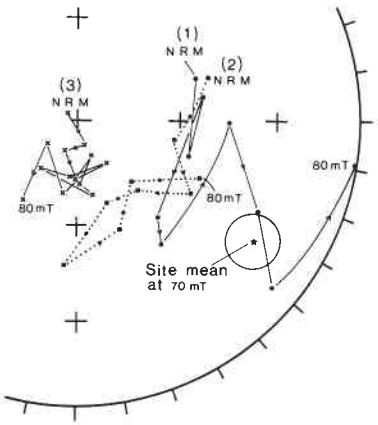


FIG. 7. Behavior of three specimens from country rock (sample 07, located 20 m from dyke margin at site 20).

Kazan dykes

Sites 16–18 and 38–40 were in northeast trending dykes, which, with the possible exception of the dyke at site 40, are thought to be members of the Kazan swarm. This swarm within the Tulemalu Lake map area is present largely in a zone 25–30 km wide along the Tulemalu Fault and is pervasively metamorphosed. Of the six sites sampled for paleomagnetic study, only sites 18 and 40 contain remnants of clinopyroxene. Sites 17 and 38 contain traces of garnet. Commonly the pyroxene has been replaced by green hornblende, which is rimmed by blue–green amphibole. This suggests two periods of alteration.

Site 16, which has two generations of amphibole, and site 18, which has clinopyroxene rimmed by amphibole, proved quite unstable in response to AF demagnetization. Site 18, however, indicated a very steep southwest component in response to thermal cleaning up to 555°C.

After AF cleaning, site 17, which is sparsely porphyritic and

TABLE 3. Paleomagnetic data for Kazan and miscellaneous sites

Site	N	Sample No.	NRM D°	NRM I°	NRM K	NRM α_{95}°	Cleaned AF (mT)	Cleaned D°	Cleaned I°	Cleaned K	Cleaned α_{95}°
					Kazan dykes						
		01	071	89			30	079	86		
		02	095	88			30	185	78		
		04	045	74			25	211	80		
16	7		052	84	94	13		186	83	94	13
		01	135	67			60	135	35		
		02	146	58			60	150	58		
		03	106	76			60	116	64		
		05	124	45			60	121	41		
		06	133	53			60	139	53		
17	6		131	60	36	13		133	51	32	14
		03	179	70			25	200	66		
		04	351	78			60	98	70		
		05	114	74			30	245	78		
		06	30	79			60	228	87		
18	6		104	85	25	19		183	82	23	20
		01	157	73			80	152	50		
		02	129	61			80	129	54		
		03	150	79			80	127	59		
		04	126	70			80	119	60		
		05	132	61			80	134	56		
		07	132	74			80	123	62		
		08	326	−34			80	098	47		
38	8			Random			126	57	53	08	
		01	156	65			60	172	32		
		02	271	69			30	267	50		
		03	178	63			70	190	42		
		05	271	76			30	266	47		
		06	201	69			40	210	70		
39	6		205	74	21	17		218	56	07	32
		01	156	74			40	214	73		
		02	190	82			40	170	81		
		03	234	78			40	278	74		
		05	129	84			40	232	61		
		07	179	82			40	250	50		
40	5		179	82	99	08		238	70	25	16
Mean direction of 6 sites above (cleaned)								171	72	13	19
Mean pole from mean direction							94.2°W	30.7°N	δm = 35°		δp = 31°
Mean pole from 6 site poles							96.4°W	32.0°N		06	31
					Miscellaneous dykes						
36	7		034	72	46	09	25	085	72	28	12
37	7			Random			25	251	−59	11	21

NOTES: N = number of cores collected at a site; site mean data are given to right of the site numbers; other symbols are the same as for Table 2.

contains traces of garnet, and site 38, which also contains garnet and two generations of hornblende, have stable southeast-down components steeper than the mean Tulemalu component but no steeper than some Tulemalu site directions. Two specimens of site 17 exhibited a much steeper southeast component in response to thermal cleaning, and two of three thermally cleaned specimens from site 38 also showed this steep southeast component. Site 40 split into two distinct groups during AF cleaning. Two specimens moved to a good Tulemalu direction and remained there even after the AF cleaning was followed by step heating to 560°C. The dyke at site 40 is megascopically similar to east-southeast Tulemalu dykes.

Site 39 was obtained from a completely recrystallized porphyritic dyke, which is megascopically similar to the Kaminak dykes (Christie *et al.* 1975). During AF cleaning two specimens indicated a stable south 40° down direction, whereas the other specimens were less stably magnetized in a westerly down direction. Thermal cleaning of a specimen from core 1 of this site removed a present Earth's field component to 400°C and above this a south 40° down component.

Paleomagnetic results for the more stable Kazan samples are given in Table 3.

Discussion

In the preceding section we have discussed some aspects of the paleomagnetism of the Tulemalu and Kazan dykes and probable age of these magnetizations. We wish now to address the broader problems of the possible correlation of these dykes

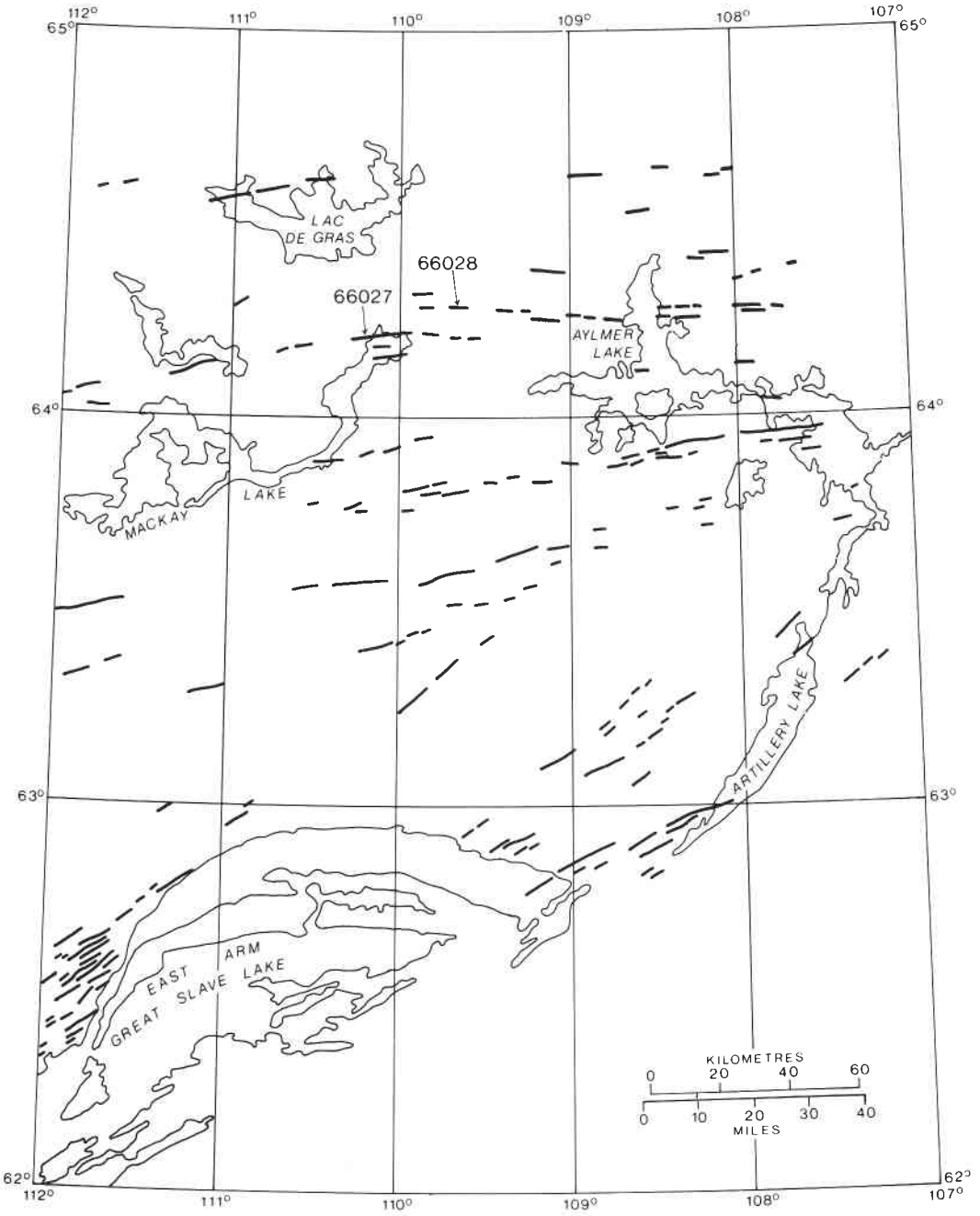


FIG. 8. Easterly trending late Archean or early Aphebian dykes in the southeast Slave Province and locations of two sites on "X" dykes (metamorphosed MacKay dyke sites 27 and 28 of Table 4).

with dyke swarms elsewhere in the Canadian Shield, the position of the Kazan and Tulemalu paleomagnetic poles relative to the polar wandering path for this part of the shield, and the possible significance of the path relative to paths of similar age elsewhere in the shield.

The Kazan dykes most closely resemble the Kaminak dykes (Chrisite *et al.* 1975) in trend, appearance, and apparent age. Both sets are sparsely porphyritic. Twelve K−Ar whole-rock ages from chilled marginal samples of eight Kaminak dykes were discussed by Christie *et al.* (1975). Five samples from three fresh dykes yielded ages greater than 2250 Ma with an average of 2370 Ma, and on the basis of this it was suggested that these dykes were intruded more than 2300 Ma ago. These whole-rock ages provide a minimum for the age of intrusion so it is likely that these dykes are Archean, i.e., greater than 2500 Ma in age. It is also likely that the magnetization of the unaltered Kaminak dykes is Archean. The above described similarities between the Kazan and Kaminak dykes suggest that these are parts of the same late Archean swarm and that the Kaminak (unmetamorphosed) pole is of late Archean age.

Dykes of trend and age similar to those of the Tulemalu swarm have been known from the Slave Province and from the Superior Province of northern Quebec for more than 20 years (Burwash *et al.* 1963; swarms 2 and 3 of Fahrig and Wanless 1963; Leech 1966). It was pointed out (Fahrig and Wanless 1963) that dyke swarms of about 2200 Ma appear to pass from the Slave Province through the western Churchill Province, where their continuations were thought at that time to be repre-

TABLE 4. Paleomagnetic data for the metamorphosed MacKay ("*X*") dykes

Site	Location (°N, °W)	*N*	AF (mT)	*D*°	*I*°	α_{95}°
19*	63.3,113.7	4	30–40	133	00	11
25*	64.5,111.4	3	30–40	125	−07	7
26*	63.3,113.6	4	30–40	137	−10	6
66027	64.2,110.1	5	20–90	138	25	19
66028	64.3,109.7	5	20–90	143	15	15
Mean direction of 5 sites (cleaned)				135	06	16
Mean pole from mean direction			115.6°E 16°N	$\delta m = 16°$	$\delta p = 8°$	
Mean pole from 5 site poles			115.6°E 16°N			09.2

NOTES: AF is range of peak alternating fields; remaining symbols as in Table 2.

*These sites are from McGlynn and Irving (1975).

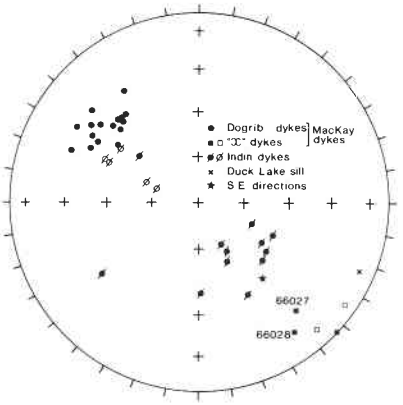


FIG. 9. Site magnetization directions for basic intrusive rocks in the southern Slave Province. All directions except 66027 and 66028 are from McGlynn and Irving (1975). Note that the "*X*" dykes are thought to be Dogrib dykes that contain an overprint with an inclination similar to the southeast directions. Open symbols indicate a north-seeking up direction.

sented entirely by metamorphosed segments, and into the Superior Province of northern Quebec.

It has been suggested that there are actually two distinct easterly trending swarms in the southern Slave Province that have yielded K–Ar ages greater than 2 Ga. Both have trends roughly parallel to the axis of the East Arm Rift and may be related to the initiation of this feature. One of these swarms is the Dogrib dykes of McGlynn and Irving (1975), which according to them contain olivine. The second swarm was first distinguished from the Dogrib by McGlynn and Irving because of its distinctive paleomagnetic signature. They further observed that these dykes lack olivine and are slightly altered. McGlynn and Irving reported results from three sites of these dykes, which they referred to as "*X*" dykes. We have data on two other dykes that are unquestionably correlatives of the "*X*" dykes, and the data for these are presented along with the earlier data in Table 4 and are located in Fig. 8. When our data were first analyzed along with the earlier data we were inclined to agree with the suggestion that the Dogrib and "*X*" dykes were two separate swarms. However, most and probably all of the "*X*" dyke samples contain two stable components, one southeast up (the "*X*" direction) and the other about 45° down. In some cases one part of a core was dominated by the *X* direction and the other by the southeast-down direction.

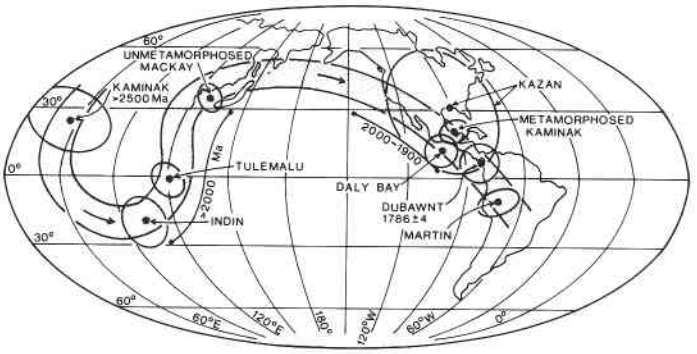


FIG. 10. Polar wandering curve for the period 2500–1800 Ma incorporating poles from rocks in the southern Churchill and southern Slave provinces. Note that the Dubawnt and Martin poles may lie on a younger loop rather than on a simple continuation of the path containing the Daly Bay, metamorphosed Kaminak (Evans and Bingham 1973), and Kazan poles.

McGlynn and Irving also observed this southeast-down direction (which they called "southeast directions") particularly in borders of some Dogrib dykes. They correctly interpreted this as a later superimposed component. The response of the "*X*" dykes to demagnetization and their low-grade metamorphism make it probable that they contain a significant southeast-down component that could not be completely removed. Similarly, it is probable that the Dogrib has on average been overprinted by this later southeast magnetization, and the Dogrib mean should be less steep to the northwest (Fig. 9). The ovals of confidence of the Dogrib and "*X*" dykes will intersect if we remove an assumed southeast overprint that is 25% of the cleaned magnetization. The mean magnetization of the Dogrib plus "*X*" would then be 311°, 19°. The pole position would be at 122°E, 27°N. This is closer to that of the Tulemalu (Fig. 10).

Considering the low-grade alteration of the "*X*" dykes and their response to demagnetization it seems logical to interpret the "*X*" dykes simply as reversed Dogrib from which the secondary southeast-down component could not be completely removed. A stumbling block to this interpretation appears to be the apparent difference in chemistry of the two swarms as indicated by the presence of olivine in only the Dogrib. There is, however, no significant difference in their chemistry, and differences in mineralogy probably relate to metamorphism. We suggest that the two swarms be referred to collectively as the MacKay dykes after MacKay Lake near which they are well exposed. The pole for the metamorphosed MacKay ("*X*") dykes is not shown in Fig. 10 because it is clearly a composite. It is noted that overprints have been reported in sedimentary strata that occupy basins adjacent to the Slave Province (Evans and Hoye 1981).

The position of early Aphebian and late Archean poles for the western Churchill Province relative to poles of similar age for the Superior Province has been briefly discussed by Christie *et al.* (1975). A comparison of poles for the two provinces provided no compelling evidence for large-scale movements between them. However, given the scatter of poles and uncertainty regarding their ages, this does not rule out large-scale relative movements subsequently followed by movement that returned the plates to their approximate original positions. One feels intuitively that more than one cycle of this type is unlikely to result in the present rather concordant configuration. In a similar way McGlynn and Irving (1975) have compared late Archean and Aphebian polar data from the Slave and Superior

provinces. They point out that the Slave and Superior provinces may have occupied approximately the same relative positions during the late Archean and early Proterozoic as they do today.

We wish now to compare the late Archean and early Proterozoic poles of the western Churchill Province with those from the Slave Province. As previously pointed out, Churchill poles of this age range fall into two distinct groups. One group of poles, including those from the Daly Bay, Kazan, and altered Kaminak dykes, is clearly the younger. It is derived from metamorphosed material that yields K−Ar ages that suggest magnetization between 1900 and 2000 Ma. The direction of younging along the section of the Churchill polar wandering curve that includes these poles is indicated by the Dubawnt volcanics, which are 1786 ± 4 Ma old (Park *et al.* 1973; Donaldson in Wanless and Loveridge 1972) and lie on the uplifted and eroded Hudsonian Orogen. The second group of poles is thought to be greater than 2200 Ma in age and comprises the Tulemalu and unaltered Kaminak dykes. They are probably more than 2500 Ma old.

Within the Slave Province the MacKay and Indin dykes represent basic intrusions that are probably greater than 2200 Ma in age (McGlynn and Irving 1975). Reasons for correlating the MacKay dykes of the Slave Province with the Tulemalu dykes of the Churchill Province have already been outlined, and the relationship of their poles is indicated in Fig. 10. We note again that the MacKay (Dogrib) probably contains an overprint that has moved its pole away from that of the Tulemalu. The MacKay dykes may be Archean in age (Gates and Hurly 1973) and have yielded a pole to the north of the Tulemalu poles. There are no known cross-cutting relationships between the MacKay and Indin dykes and the relative ages of their magnetization are unknown but the path is shown as younging northward from the Indin to MacKay to tie in with the path of appropriate age as shown by Schwarz and Freda (1983).

BURWASH, R. A., BAADSGAARD, H., CAMPBELL, F. A., DUMMING, G. L., and FOLINSBEE, R. E. 1963. Potassium−argon dates of diabase dyke-systems. District of Mackenzie, N.W.T. Transactions of the Canadian Institute of Mining and Metallurgy, **66**, pp. 303−307.

CHRISTIE, K. W., DAVIDSON, A., and FAHRIG, W. F. 1975. The paleomagnetism of Kaminak dikes—no evidence of significant Hudsonian plate motion. Canadian Journal of Earth Sciences, **12**, pp. 2048−2064.

EADE, K. E. 1984. Geology of Tulemalu Lake and of Ferguson Lake west half, District of Keewatin. Geological Survey of Canada, paper. (In press.)

EADE, K. E., LOVERIDGE, W. D., SULLIVAN, R. W., and STEVENS, R. D. 1984. U−Pb zircon ages and Rb−Sr isotopic studies of rocks from the vicinity of Edchon and Yathkyed lakes, southern District of Keewatin. *In* Rb−Sr and U−Pb isotopic age studies, report 6. Geological Survey of Canada, paper. (In press.)

EVANS, M. E., and BINGHAM, D. K. 1973. Paleomagnetism of the Precambrian Martin Formation, Saskatchewan. Canadian Journal of Earth Sciences, **10**, pp. 1485−1493.

EVANS, M. E., and HOYE, G. S. 1981. Paleomagnetic results from the lower Proterozoic rocks of Great Slave Lake and Bathurst Inlet area, Northwest Territories. *In* Proterozoic basins of Canada. *Edited by* F. H. A. Campbell. Geological Survey of Canada, Paper 81-10, pp. 191−202.

FAHRIG, W. F., and WANLESS, R. K. 1963. Age and significance of diabase dyke swarms of the Canadian Shield. Nature, **200**, pp. 934−937.

GATES, T. M., and HURLEY, P. M. 1973. Evaluation of Rb−Sr dating methods applied to the Matachewan, Abitibi, Mackenzie, and Sudbury dike swarms in Canada. Canadian Journal of Earth Sciences, **10**, pp. 900−919.

LEECH, A. P. 1966. Potassium−argon dates of intrusive rocks of the District of MacKenzie, N.W.T. Canadian Journal of Earth Sciences, **3**, pp. 389−412.

MCGLYNN, J. C., and IRVING, E. 1975. Paleomagnetism of early Aphebian diabase dykes from the Slave Structural Province, Canada. Tectonophysics, **26**, pp. 23−38.

PARK, J. K. 1973. Paleomagnetism of metamorphic rocks of Daly Bay and Melville Peninsula, Churchill Structural Province. Canadian Journal of Earth Sciences, **10**, pp. 1079−1088.

PARK, J. K., IRVING, E., and DONALDSON, J. A. 1973. Paleomagnetism of the Precambrian Dubawnt Group. Geological Society of America Bulletin, **84**, pp. 859−870.

SCHWARZ, E. J., and FREDA, G. N. 1983. Paleomagnetism and time-stratigraphic correlation of Proterozoic redbeds of the Labrador Trough and outliers of northern Quebec. Canadian Journal of Earth Sciences, **20**, pp. 1725−1737.

STOCKWELL, C. H. 1982. Proposals for time classification and correlation of Precambrian rocks and events in Canada and adjacent areas of the Canadian Shield. Part 1: A time classification of Precambrian rocks and events. Geological Survey of Canada, Paper 80-19.

WANLESS, R. K., and LOVERIDGE, W. D. 1972. Rubidium−strontium isochron age studies, report 1. Geological Survey of Canada, Paper 72-23, pp. 25−31.

WANLESS, R. K., STEVENS, R. D., LACHANCE, G. R., and DELABIO, R. N. 1980. Age determinations and geological studies, K−Ar isotopic ages, report 14. Geological Survey of Canada, Paper 79-2, pp. 40−46.